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Least squares solutions to the equations $AX = B,\ XC = D$ with some constraints

Yuyang Qiu [a,*,1], Anding Wang [b]

[a] College of Statistics and Mathematics, Zhejiang Gongshang University, Hangzhou 310018, PR China
[b] College of Information and Electronics, Zhejiang Gongshang University, Hangzhou 310018, PR China

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ABSTRACT

The least squares solutions to the equations $AX = B,\ XC = D$ with some constraints such as orthogonality, symmetric orthogonality, symmetric idempotence are considered. By one time singular value decomposition or eigenvalue decomposition of the matrix product generated by the coefficient matrices $A,\ C$ and right hand side matrices $B,\ D$, the constrained solutions are constructed simply. Similar strategy is applied to the equations with corresponding P-commuting constraints with given symmetric matrix P. Numerical examples that show the efficiency of the proposed methods are presented.

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1. Introduction

Given matrices $A,\ B \in \mathscr{R}^{m\times n}$, $C,\ D \in \mathscr{R}^{n\times p}$, and matrix constrained space $\mathscr{S} \subset \mathscr{R}^{n\times n}$ possessing special structures such as orthogonality, symmetric orthogonality, symmetric idempotence, and their corresponding P-commuting constraints with given symmetric matrix P, we consider the least squares solutions to the following matrix equations:

$$AX = B, \quad XC = D \tag{1.1}$$

with $\mathscr{S}$ constraint.

Eq. (1.1) with or without constraints, together with their corresponding least squares problem, have been discussed by many scholars recently [1,10,15,16]. Necessary and sufficient conditions for the existence of a solution to the equations are presented and general solutions are given in terms of generalized inverse in [1,10,16]. For Eq. (1.1) with $PX = sXP$ constraint, Zhang also has proposed an eigenvector-free formula of the solutions in [15], where P is a given Hermitian matrix satisfying $P^2 = I$ and $s = \pm 1$. And, their special case: $AX = B$ and its least squares problem (Procrustes problem) have wide applications in the engineering and many other domains, more and more people have been attracted in this topic [5,7,8,14]. By several times singular value decompositions (SVDs), CS decompositions and eigenvalue decompositions, Hu and his collaborators have presented the detailed solutions to Procrustes problem and its corresponding equation with symmetric orthogonal, skew-symmetric orthogonal, reflexive, or anti-reflexive constraints [7,8,14]. Moreover, in terms of generalized singular value decomposition (GSVD) [12,13] and the canonical correlation decomposition (CCD) [4] of the coefficient matrices, the extended problem of (1.1): $A_iXB_i = C_i$, $i = 1, 2$ has been considered and some well-known results have been proposed in [2,6,9,11]. However, the involved decompositions in both the special case and the extended one are always intricate, the representation forms of the solutions are complex, and numerical difficulty may occur. So they may not be suitable for the constrained problem (1.1) and seeking for a simpler way will be necessary.

* Corresponding author.
E-mail addresses: YuyangQiu@gmail.com (Y. Qiu), anding_704@hotmail.com (A. Wang).
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Since,

$$\left\|\begin{pmatrix} AX-B \\ C^{\mathrm{T}}X^{\mathrm{T}}-D^{\mathrm{T}}\end{pmatrix}\right\|_F^2 = \operatorname{trace}\left((X^{\mathrm{T}}A^{\mathrm{T}}-B^{\mathrm{T}}, XC-D)\begin{pmatrix} AX-B \\ C^{\mathrm{T}}X^{\mathrm{T}}-D^{\mathrm{T}}\end{pmatrix}\right) = \operatorname{trace}(XX^{\mathrm{T}}A^{\mathrm{T}}A + X^{\mathrm{T}}XCC^{\mathrm{T}} - 2X^{\mathrm{T}}(A^{\mathrm{T}}B+DC^{\mathrm{T}})) + \|B\|_F^2 + \|D\|_F^2, \tag{1.2}$$

then we can ask for the least squares solutions to (1.1) with further constraint $\mathscr{S}$ only by minimizing (1.2) with the same constraint. Note that Golub and Van Loan have presented a simple method to construct the solution of the orthogonal balanced Procrustes problem: they transform the original problem to minimize the trace composed by a linear mapping of unknown matrix X, and obtain the solutions explicitly by one time SVD [3, p. 601]. Their idea is novel and efficient, and it is suitable for the constrained least squares problem of (1.1). Our ideas base on the following facts: (1) for the special constraints such as orthogonality, symmetric orthogonality, and symmetric idempotence, the quadratic function in (1.2) can be simplified as a linear one; (2) the solutions to the linear equivalent form are obtained by one time SVD or eigenvalue decomposition of the matrix product generated by the coefficient matrices A, C and right hand side matrices B, D. That method can also be generalized to the corresponding P-commuting orthogonal, symmetric orthogonal, or symmetric idempotent constrained least squares problems with given symmetric matrix P.

The rest of paper is organized as follows. In Section 2, we consider the least squares solutions to Eq. (1.1) with orthogonal constraint. For the other two special constraints, that is, symmetric orthogonality and symmetric idempotence, similar conclusions are also proposed in Sections 3 and 4. The corresponding P-commuting orthogonal, symmetric orthogonal, or symmetric idempotent constrained least squares problems are discussed in Section 5. Numerical examples are given in Section 6 to display the efficiency of the algorithms.

Notation. Let $\mathscr{R}^{m\times n}$ denote the space of real $m\times n$ matrix. For any matrix $X=(x_{ij})\in\mathscr{R}^{m\times n}$, $\operatorname{trace}(X)$ is its trace, $\|\cdot\|_F$ is the Frobenius norm of matrix. If X is a square matrix with order n, then $\operatorname{sym}(X)$ stands for the symmetric part $\operatorname{sym}(X)=\frac{1}{2}(X+X^{\mathrm{T}})$, $D_X=\operatorname{diag}(x_{11},\ldots,x_{nn})$ is the diagonal matrix composed by the diagonal elements of X. We also denote by $\mathrm{O}\mathscr{R}^n$, I_n the orthogonal matrix space and identity matrix with order n, respectively. And $\mathrm{O}_{m\times n}$ refers to $m\times n$ zero matrix, O_n is the zero matrix with order n. Meanwhile, $\mathscr{X}_{\mathscr{S}}$ denotes the solutions set to the least squares problem of Eq. (1.1) with $\mathscr{S}$ constraint, that is,

$$\mathscr{X}_{\mathscr{S}} = \left\{X : X = \arg\min_{X\in\mathscr{S}} \left\|\begin{pmatrix} AX-B \\ C^{\mathrm{T}}X^{\mathrm{T}}-D^{\mathrm{T}}\end{pmatrix}\right\|_F\right\},$$

where arg is the value of the argument.

2. Least squares solutions to (1.1) with orthogonal constraint

Let $\mathscr{S}$ be the orthogonal matrix space with order n, that is, $\mathscr{S}=\mathrm{O}\mathscr{R}^n$, (1.2) is equivalent to,

$$\|A\|_F^2 + \|B\|_F^2 + \|C\|_F^2 + \|D\|_F^2 - 2\operatorname{trace}(X^{\mathrm{T}}(A^{\mathrm{T}}B+DC^{\mathrm{T}})). \tag{2.1}$$

Denote by $W=A^{\mathrm{T}}B+DC^{\mathrm{T}}$, then minimizing (1.2) is equivalent to maximizing $\operatorname{trace}(X^{\mathrm{T}}W)$. Hence,

$$\mathscr{X}_{\mathscr{S}} = \left\{X : X = \arg\max_{X\in\mathscr{S}} \operatorname{trace}(X^{\mathrm{T}}W)\right\}.$$

Let SVD of W be

$$W = U\operatorname{diag}(\Sigma_r, \mathrm{O}_{n-r})V^{\mathrm{T}}, \tag{2.2}$$

where

$$U,\ V\in\mathrm{O}\mathscr{R}^n, \quad \Sigma_r=\operatorname{diag}(\sigma_1,\ldots,\sigma_r), \quad 0<\sigma_r\leqslant\cdots\leqslant\sigma_1,\ \ r=\operatorname{rank}(W).$$

Set $\widetilde{X}=U^{\mathrm{T}}XV$,

$$\operatorname{trace}(X^{\mathrm{T}}W) = \operatorname{trace}(\widetilde{X}^{\mathrm{T}}\operatorname{diag}(\Sigma_r,\mathrm{O}_{n-r})).$$

So,

$$\mathscr{X}_{\mathscr{S}} = \left\{X : X = U\widetilde{X}V^{\mathrm{T}},\ \widetilde{X} = \arg\max_{\widetilde{X}\in\mathscr{S}} \operatorname{trace}(\widetilde{X}^{\mathrm{T}}\operatorname{diag}(\Sigma_r,\mathrm{O}_{n-r}))\right\}. \tag{2.3}$$

Partitioning

$$\widetilde{X} = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22}\end{pmatrix}, \quad X_{11}\in\mathscr{R}^{r\times r},\ X_{22}\in\mathscr{R}^{(n-r)\times(n-r)},$$

then

$$\max_{\widetilde{X}\in\mathscr{S}} \operatorname{trace}(\widetilde{X}^{\mathrm{T}}\operatorname{diag}(\Sigma_r,\mathrm{O}_{n-r})) = \max_{X_{11}} \operatorname{trace}(X_{11}^{\mathrm{T}}\Sigma_r). \tag{2.4}$$

Note that $\widetilde{X} = (\tilde{x}_{ij})$ is also orthogonal, then

$$-1 \leqslant \tilde{x}_{ij} \leqslant 1, \quad i, j = 1, \ldots, n.$$

So (2.4) implies:

$$D_{X_{11}} = I_r.$$

Since the columns and rows of $\widetilde{X}$ are orthonormal,

$$X_{11} = I_r, \quad X_{12} = \mathrm{O}_{r\times n-r}, \quad X_{21} = \mathrm{O}_{n-r\times r}, \quad X_{22} \in \mathrm{O}\mathscr{R}^{n-r} \text{ is arbitrary.}$$

Hence,

$$\mathscr{X}_{\mathscr{S}} = \left\{ X : X = U \begin{pmatrix} I_r & \\ & G \end{pmatrix} V^{\mathrm{T}} \right\}, \tag{2.5}$$

where $G \in \mathrm{O}\mathscr{R}^{n-r}$ is arbitrary. Above analysis leads to the following theorem.

Theorem 2.1. *Denote by $W = A^{\mathrm{T}}B + DC^{\mathrm{T}}$, and let SVD of W be (2.2). Then the least squares solutions to Eq. (1.1) with orthogonal constraint are*

$$X = U \begin{pmatrix} I_r & \\ & G \end{pmatrix} V^{\mathrm{T}},$$

where the matrix $G \in \mathrm{O}\mathscr{R}^{n-r}$ is arbitrary.

3. Least squares solutions to (1.1) with symmetric orthogonal constraint

In this case, $\mathscr{S} = \{X : X^{\mathrm{T}} = X,\ X \in \mathrm{O}\mathscr{R}^n\}$. Denote by $W = A^{\mathrm{T}}B + DC^{\mathrm{T}}$, together with X is symmetric,

$$\mathrm{trace}(XW) = \mathrm{trace}(XW^{\mathrm{T}}) = \mathrm{trace}(X\,\mathrm{sym}(W)).$$

So,

$$\mathscr{X}_{\mathscr{S}} = \left\{ X : X = \arg\max_{X\in\mathscr{S}} \mathrm{trace}(X\,\mathrm{sym}(W)) \right\}.$$

Let an eigenvalue decomposition of the matrix $\mathrm{sym}(W)$ be

$$\mathrm{sym}(W) = U\,\mathrm{diag}(\lambda_1 I_{l_1}, \ldots, \lambda_t I_{l_t})U^{\mathrm{T}}, \tag{3.1}$$

where

$$U \in \mathrm{O}\mathscr{R}^n, \quad \lambda_1 < \cdots < \lambda_{s-1} < \lambda_s (= 0) < \lambda_{s+1} < \cdots < \lambda_t, \quad \sum_{j=1}^{t} l_j = n.$$

Set $\widetilde{X} = U^{\mathrm{T}}XU$, then $\widetilde{X} \in \mathscr{S}$ and

$$\mathscr{X}_{\mathscr{S}} = \left\{ X : X = U\widetilde{X}U^{\mathrm{T}},\ \widetilde{X} = \arg\max_{\widetilde{X}\in\mathscr{S}} \mathrm{trace}(\widetilde{X}\,\mathrm{diag}(\lambda_1 I_{l_1}, \ldots, \lambda_t I_{l_t})) \right\}.$$

Partition $\widetilde{X} = (X_{ij})$ with $X_{ij} \in \mathscr{R}^{l_i\times l_j}$,

$$\mathrm{trace}(\widetilde{X}\,\mathrm{diag}(\lambda_1 I_{l_1}, \ldots, \lambda_t I_{l_t})) = \sum_{i=1}^{t} \mathrm{trace}(\lambda_i X_{ii}). \tag{3.2}$$

Since $X_{ii},\ i = 1, \ldots, t$ are independent,

$$\max_{\widetilde{X}} \mathrm{trace}(\widetilde{X}\,\mathrm{diag}(\lambda_1 I_{l_1}, \ldots, \lambda_t I_{l_t})) = \sum_{i=1}^{t} \max_{X_{ii}} \mathrm{trace}(\lambda_i X_{ii}). \tag{3.3}$$

Note that the matrix $\widetilde{X}$ is orthogonal, its element $\tilde{x}_{ij}$ satisfies

$$-1 \leqslant \tilde{x}_{ij} \leqslant 1, \quad i, j = 1, \ldots, n.$$

Together with (3.3),

$$D_{X_{ii}} = \begin{cases} -I_{l_i} & 1 \leqslant i \leqslant s-1, \\ I_{l_i} & s+1 \leqslant i \leqslant t, \end{cases}$$

and subblock X_{ss} corresponding to zero eigenvalue is arbitrary. Denote by $n_j = \sum_{i=1}^{j} l_i$, together with the symmetry, $\widetilde{X}$ also can be partitioned as

$$\widetilde{X} = \begin{pmatrix} \widetilde{X}_{11} & \widetilde{X}_{12} & \widetilde{X}_{13} \\ \widetilde{X}_{12}^{\mathrm{T}} & G_{l_s} & \widetilde{X}_{23} \\ \widetilde{X}_{13}^{\mathrm{T}} & \widetilde{X}_{23}^{\mathrm{T}} & \widetilde{X}_{33} \end{pmatrix},$$

where $G_{l_s} \in \mathscr{R}^{l_s \times l_s}$ is symmetric, $\widetilde{X}_{11} \in \mathscr{R}^{n_{s-1} \times n_{s-1}}$ and $\widetilde{X}_{33} \in \mathscr{R}^{n-n_s \times n-n_s}$ satisfy

$$D_{\widetilde{X}_{11}} = -I_{n_{s-1}}, \quad D_{\widetilde{X}_{33}} = I_{n-n_s}.$$

With the orthogonality of $\widetilde{X}$, it is not difficult to verify:

$$\widetilde{X}_{11} = -I_{n_{s-1}}, \quad \widetilde{X}_{12} = \mathrm{O}_{n_{s-1} \times l_s}, \quad \widetilde{X}_{13} = \mathrm{O}_{n_{s-1} \times n-n_s}, \quad \widetilde{X}_{23} = \mathrm{O}_{l_s \times n-n_s}, \quad \widetilde{X}_{33} = I_{n-n_s}$$

and G_{l_s} is also symmetric orthogonal. Thus, we have the following theorem.

Theorem 3.1. *Denote by $W = A^{\mathrm{T}}B + DC^{\mathrm{T}}$, and let an eigenvalue decomposition of the matrix* sym(W) *be (3.1). Then the least squares solutions to Eq. (1.1) with symmetric orthogonal constraint are*

$$X = U \begin{pmatrix} -I_{n_{s-1}} & & \\ & G_{l_s} & \\ & & I_{n-n_s} \end{pmatrix} U^{\mathrm{T}}, \tag{3.4}$$

where $G_{l_s} \in \mathscr{R}^{l_s \times l_s}$ is arbitrary symmetric orthogonal, and $n_j = \sum_{i=1}^{j} l_i$.

4. Least squares solutions to (1.1) with symmetric idempotent constraint

Let constraint be symmetric idempotence, that is,

$$\mathscr{S} = \{X : X^2 = X,\ X^{\mathrm{T}} = X,\ X \in \mathscr{R}^{n \times n}\},$$

(1.2) is also rewritten as

$$\mathrm{trace}(XW) + \|B\|_F^2 + \|D\|_F^2 \tag{4.1}$$

with

$$W = A^{\mathrm{T}}A + CC^{\mathrm{T}} - 2(A^{\mathrm{T}}B + DC^{\mathrm{T}}).$$

Therefore,

$$\mathscr{X}_{\mathscr{S}} = \left\{ X : X = \arg\min_{X \in \mathscr{S}} \mathrm{trace}(X\,\mathrm{sym}(W)) \right\}. \tag{4.2}$$

Let an eigenvalue decomposition of the symmetric matrix sym(W) be (3.1), and denote by $\widetilde{X} = U^{\mathrm{T}}XU$,

$$\mathscr{X}_{\mathscr{S}} = \left\{ X : X = U\widetilde{X}U^{\mathrm{T}},\ \widetilde{X} = \arg\min_{\widetilde{X} \in \mathscr{S}} \mathrm{trace}(\widetilde{X}\,\mathrm{diag}(\lambda_1 I_{l_1}, \ldots, \lambda_t I_{l_t})) \right\}. \tag{4.3}$$

We also partition $\widetilde{X} = (X_{ij})$ with $X_{ij} \in \mathscr{R}^{l_i \times l_j}$,

$$\min_{\widetilde{X} \in \mathscr{S}} \mathrm{trace}(\widetilde{X}\,\mathrm{diag}(\lambda_1 I_{l_1}, \ldots, \lambda_t I_{l_t})) = \sum_{i=1}^{t} \min_{X_{ii}} \mathrm{trace}(\lambda_i X_{ii}). \tag{4.4}$$

Note that $\widetilde{X} = (\tilde{x}_{ij})$ is also symmetric idempotent, so $0 \leqslant \tilde{x}_{ii} \leqslant 1,\ i = 1, \ldots, n$. With (4.4), the subblocks $X_{ii},\ i = 1, \ldots, t$ satisfy

$$D_{X_{ii}} = \begin{cases} I_{l_i} & 1 \leqslant i \leqslant s-1, \\ \mathrm{O}_{l_i} & s+1 \leqslant i \leqslant t, \end{cases}$$

and X_{ss} is an arbitrary symmetric matrix. Denote by $n_j = \sum_{i=1}^{j} l_i$, we write $\widetilde{X}$ as

$$\widetilde{X} = \begin{pmatrix} \widetilde{X}_{11} & \widetilde{X}_{12} & \widetilde{X}_{13} \\ \widetilde{X}_{12}^{\mathrm{T}} & G_{l_s} & \widetilde{X}_{23} \\ \widetilde{X}_{13}^{\mathrm{T}} & \widetilde{X}_{23}^{\mathrm{T}} & \widetilde{X}_{33} \end{pmatrix}$$

with $\widetilde{X}_{11} \in \mathscr{R}^{n_{s-1} \times n_{s-1}}$, $\widetilde{X}_{33} \in \mathscr{R}^{n-n_s \times n-n_s}$, and $G_{l_s} \in \mathscr{R}^{l_s \times l_s}$. It is obvious that

$$D_{\widetilde{X}_{11}} = I_{n_{s-1}}, \quad D_{\widetilde{X}_{33}} = \mathrm{O}_{n-n_s}. \tag{4.5}$$

With the symmetric idempotence $\widetilde{X}^2 = \widetilde{X}$, the first equality of (4.5) leads to

$$\widetilde{X}_{11} = I_{n_{s-1}}, \quad \widetilde{X}_{12} = \mathrm{O}_{n_{s-1}\times l_s}, \quad \widetilde{X}_{13} = \mathrm{O}_{n_{s-1}\times n-n_s},$$

and the last one implies

$$\widetilde{X}_{23} = \mathrm{O}_{l_s\times n-n_s}, \quad \widetilde{X}_{33} = \mathrm{O}_{n-n_s}.$$

Therefore, the following theorem holds.

Theorem 4.1. *Denote by $W = A^{\mathrm{T}}A + CC^{\mathrm{T}} - 2(A^TB + DC^T)$, and let an eigenvalue decomposition of* sym(W) *be (3.1). Then the least squares solutions to Eq. (1.1) with symmetric idempotent constraint are*

$$X = U\begin{pmatrix} I_{n_{s-1}} & & \\ & G_{l_s} & \\ & & \mathrm{O}_{n-n_s} \end{pmatrix}U^{\mathrm{T}}, \tag{4.6}$$

where $G_{l_s} \in \mathscr{R}^{l_s\times l_s}$ is arbitrary symmetric idempotent, and $n_j = \sum_{i=1}^{j} l_i$.

5. Least squares solutions to (1.1) with corresponding *P*-commuting constraint

In this section, we generalize the constraints to their corresponding P-commuting constraints, where P is a given symmetric matrix. Let an eigenvalue decomposition of P be

$$P = H\,\mathrm{diag}(\tilde{\lambda}_1 I_{k_1}, \ldots, \tilde{\lambda}_p I_{k_p})H^{\mathrm{T}}, \tag{5.1}$$

where $H \in \mathrm{O}\mathscr{R}^n$. A matrix X is commuting to P, that is, $PX = XP$, if and only if

$$X = H\,\mathrm{diag}(X_1, \ldots, X_p)H^{\mathrm{T}}, \tag{5.2}$$

where $X_i \in \mathscr{R}^{k_i\times k_i}$. Clearly, $\{X_i\}_{i=1}^{p}$ are the independent parameter matrices if no other constraints are imposed. In the following discussion, we always assume that the decomposition (5.1) holds.

If X is further required to be orthogonal, symmetric orthogonal, or symmetric idempotent, then all the X_i in (5.2) should satisfy the same constraints, too. In order to discuss the least squares solutions to (1.1) with different P-commuting constraints, we introduce the following lemma.

Lemma 5.1. *Partition the n order matrix $W = (W_{ij})$ conforming to (5.2), that is, subblock $W_{ij} \in \mathscr{R}^{k_i\times k_j}$, then*

$$\mathrm{trace}(\mathrm{diag}(X_1, \ldots, X_p)W) = \sum_{j=1}^{p} \mathrm{trace}(X_i W_{ii}).$$

Proof. Note that

$$\mathrm{diag}((X_1, \ldots, X_p)W) = \begin{pmatrix} X_1W_{11} & X_1W_{12} & \ldots & X_1W_{1p} \\ X_2W_{21} & X_2W_{22} & \ldots & X_2W_{2p} \\ \ldots & \ldots & \ldots & \ldots \\ X_pW_{p1} & X_2W_{p2} & \ldots & X_pW_{pp} \end{pmatrix},$$

so,

$$\mathrm{trace}(\mathrm{diag}(X_1, \ldots, X_p)W) = \sum_{j=1}^{p} \mathrm{trace}(X_i W_{ii}). \qquad \square$$

5.1. The P-commuting orthogonal constraint

In this case,

$$\mathscr{S} = \{X : PX = XP,\ X \in \mathrm{O}\mathscr{R}^n\}.$$

Denote by $\widetilde{W} = H^{\mathrm{T}}WH$ with $W = A^{\mathrm{T}}B + DC^{\mathrm{T}}$, and partition $\widetilde{W}$ by (W_{ij}) with $W_{ij} \in \mathscr{R}^{k_i\times k_j}$,

$$\mathscr{X}_{\mathscr{S}} = \left\{X : X = \arg\max_{X\in\mathscr{S}} \mathrm{trace}(X^{\mathrm{T}}W)\right\} = \{X : X = H\,\mathrm{diag}(X_1, \ldots, X_p)H^{\mathrm{T}}\}$$

with

$$(X_1, \ldots, X_p) = \arg \max_{X_i\in\mathrm{O}\mathscr{R}^{k_i}, i=1,\ldots,p} \mathrm{trace}(\mathrm{diag}(X_1^{\mathrm{T}}, \ldots, X_p^{\mathrm{T}})\widetilde{W}).$$

With Lemma 5.1, together with $X_i,\ i = 1,\ldots,p$ are independent,

$$X_i = \arg\max_{X_i \in \mathrm{O}\mathscr{R}^{k_i}} \operatorname{trace}(X_i^{\mathrm{T}} W_{ii}), \quad i = 1,\ldots,p.$$

The above maximization problem can be solved as that of Theorem 2.1, so we have the following theorem.

Theorem 5.1. *Denote by $\widetilde{W} = H^{\mathrm{T}}WH$ with $W = A^{\mathrm{T}}B + DC^{\mathrm{T}}$, and partition the matrix $\widetilde{W} = (W_{ij})$ conforming to (5.2), where $W_{ij} \in \mathscr{R}^{k_i \times k_j}$. Let SVD of the matrix W_{ii} be*

$$W_{ii} = U_i \operatorname{diag}(\Sigma_{r_i}^{(i)}, \mathrm{O}_{k_i - r_i}) V_i^{\mathrm{T}}, \tag{5.3}$$

where $U_i,\ V_i \in \mathrm{O}\mathscr{R}^{k_i}$, $\Sigma_{r_i}^{(i)} = \operatorname{diag}(\sigma_1^{(i)},\ldots,\sigma_{r_i}^{(i)})$, $0 < \sigma_{r_i}^{(i)} \leqslant \cdots \leqslant \sigma_1^{(i)}$, $r_i = \operatorname{rank}(W_{ii})$, $i = 1,\ldots,p$. Then the least squares solutions to Eq. (1.1) with P-commuting orthogonal constraint are

$$X = H \operatorname{diag}(X_1,\ldots,X_p) H^{\mathrm{T}},$$

where X_i satisfies

$$X_i = U_i \begin{pmatrix} I_{r_i} & \\ & G_{k_i - r_i} \end{pmatrix} V_i^{\mathrm{T}},$$

and $G_{k_i - r_i} \in \mathrm{O}\mathscr{R}^{k_i - r_i}$ is arbitrary.

5.2. The P-commuting symmetric orthogonal constraint

Let the constraint be *P*-commuting symmetric orthogonal one, that is,

$$\mathscr{S} = \{X : X^{\mathrm{T}} = X,\ PX = XP,\ X \in \mathrm{O}\mathscr{R}^n\}.$$

Set W and $\widetilde{W}$ as in Section 5.1. Together with Lemma 5.1, we have:

$$\mathscr{X}_{\mathscr{S}} = \left\{X : X = \arg\max_{X \in \mathscr{S}} \operatorname{trace}(XW)\right\} = \{X : X = H \operatorname{diag}(X_1,\ldots,X_p) H^{\mathrm{T}}\},$$

where

$$X_i = \arg\max_{X_i = X_i^{\mathrm{T}},\, X_i \in \mathrm{O}\mathscr{R}^{k_i}} \operatorname{trace}(X_i \operatorname{sym}(W_{ii})), \quad i = 1,\ldots,p.$$

Similar to Theorem 3.1, the following theorem holds.

Theorem 5.2. *Denote by $\widetilde{W} = H^{\mathrm{T}}WH$ with $W = A^{\mathrm{T}}B + DC^{\mathrm{T}}$, and partition the matrix $\widetilde{W} = (W_{ij})$ conforming to (5.2), where $W_{ij} \in \mathscr{R}^{k_i \times k_j}$. Let an eigenvalue decomposition of the matrix $\operatorname{sym}(W_{ii})$ be*

$$\operatorname{sym}(W_{ii}) = U_i \operatorname{diag}\left(\lambda_1^{(i)} I_{l_1^{(i)}},\ldots,\lambda_{t_i}^{(i)} I_{l_{t_i}^{(i)}}\right) U_i^{\mathrm{T}}, \tag{5.4}$$

where $U_i \in \mathrm{O}\mathscr{R}^{k_i}$, $\lambda_1^{(i)} < \cdots < \lambda_{s_i-1}^{(i)} < \lambda_{s_i}^{(i)} (= 0) < \lambda_{s_i+1}^{(i)} < \cdots < \lambda_{t_i}^{(i)}$, $k_i = \sum_{j=1}^{t_i} l_j^{(i)}$, $i = 1,\ldots,p$. The least squares solutions to Eq. (1.1) with P-commuting symmetric orthogonal constraint are

$$X = H \operatorname{diag}(X_1,\ldots,X_p) H^{\mathrm{T}},$$

where X_i satisfies

$$X_i = U_i \begin{pmatrix} -I_{n_{s_i-1}^{(i)}} & & \\ & G_{l_{s_i}^{(i)}} & \\ & & I_{k_i - n_{s_i}^{(i)}} \end{pmatrix} U_i^{\mathrm{T}},$$

$G_{l_{s_i}^{(i)}}$ is an arbitrary symmetric orthogonal matrix with order $l_{s_i}^{(i)}$, and $n_j^{(i)} = \sum_{t=1}^{j} l_t^{(i)}$.

5.3. The P-commuting idempotent constraint

Similarly, for Eq. (1.1) with *P*-commuting symmetric idempotent constraint,

$$\mathscr{S} = \{X : X^{\mathrm{T}} = X,\ X^2 = X,\ PX = XP,\ X \in \mathscr{R}^{n \times n}\}.$$

Denote by $W = A^{\mathrm{T}}A + CC^{\mathrm{T}} - 2(A^{\mathrm{T}}B + DC^{\mathrm{T}})$,

$$\mathscr{X}_{\mathscr{S}} = \left\{X : X = \arg\min_{X \in \mathscr{S}} \operatorname{trace}(XW)\right\} = \{X : X = H \operatorname{diag}(X_1,\ldots,X_p) H^{\mathrm{T}}\}$$

with

$$X_i = \arg \min_{X_i = X_i^{\mathrm{T}} X_i^2 = X_i} \operatorname{trace}(X_i \operatorname{sym}(W_{ii})), \quad i = 1, \ldots, p.$$

With Theorem 4.1, the following conclusion is obvious.

Theorem 5.3. *Denote by* $\widetilde{W} = H^{\mathrm{T}} W H$ *with* $W = A^{\mathrm{T}}A + CC^{\mathrm{T}} - 2(A^{\mathrm{T}}B + DC^{\mathrm{T}})$, *and partition the matrix* $\widetilde{W} = (W_{ij})$ *conforming to (5.2), where* $W_{ij} \in \mathscr{R}^{k_i \times k_j}$. *Let an eigenvalue decomposition of the matrix* $\operatorname{sym}(W_{ii})$ *be*

$$\operatorname{sym}(W_{ii}) = U_i \operatorname{diag}\left(\lambda_1^{(i)} I_{l_1^{(i)}}, \ldots, \lambda_{t_i}^{(i)} I_{l_{t_i}^{(i)}}\right) U_i^{\mathrm{T}}, \tag{5.5}$$

where $U_i \in \mathrm{O}\mathscr{R}^{k_i}$, $\lambda_1^{(i)} < \cdots < \lambda_{s_i-1}^{(i)} < \lambda_{s_i}^{(i)} (= 0) < \lambda_{s_i+1}^{(i)} < \cdots < \lambda_{t_i}^{(i)}$, $k_i = \sum_{j=1}^{t_i} l_j^{(i)}$, $i = 1, \ldots, p$. *The least squares solutions to Eq. (1.1) with P-commuting symmetric idempotent constraint are*

$$X = H \operatorname{diag}(X_1, \ldots, X_p) H^{\mathrm{T}},$$

where X_i *satisfies*

$$X_i = U_i \begin{pmatrix} I_{n_{s_i-1}^{(i)}} & & \\ & G_{l_{s_i}^{(i)}} & \\ & & \mathrm{O}_{k_i - n_{s_i}^{(i)}} \end{pmatrix} U_i^{\mathrm{T}},$$

$G_{l_{s_i}^{(i)}} \in \mathscr{R}^{l_{s_i}^{(i)} \times l_{s_i}^{(i)}}$ *is arbitrary symmetric idempotent, and* $n_j^{(i)} = \sum_{t=1}^{j} l_t^{(i)}$.

6. Numerical example

In this section, we present some numerical examples to illustrate the efficiency of our algorithms. The coefficient matrices A and C are randomly constructed with given singular values (for simplicity, we set $m = n = p$). For example,

$$A = U \operatorname{diag}(\sigma_1, \ldots, \sigma_n) V^{\mathrm{T}} \tag{6.1}$$

with orthogonal matrices U and V constructed as follows (in MATLAB notation):

$$[U, \mathrm{temp}] = \mathrm{qr}(1 - 2\,\mathrm{rand}(n)), \quad [V, \mathrm{temp}] = \mathrm{qr}(1 - 2\,\mathrm{rand}(n)).$$

The singular values $\{\sigma_i\}$ will be chosen such that the resulting coefficient matrices A and C can have preset condition numbers. We set the residual error ϵ by

$$\epsilon = \left\| \begin{pmatrix} B - AX \\ D^{\mathrm{T}} - C^{\mathrm{T}} X^{\mathrm{T}} \end{pmatrix} \right\|_F$$

with the computational matrix X. And the right hand side matrices B, D are chosen by

$$B = AX^*, \quad D = X^* C \tag{6.2}$$

with $X^* \in \mathscr{S}$, such that the residual error of an optimal constrained least squares solution to Eq. (1.1) should be zero.

We report numerical results for the least squares solutions to Eq. (1.1) with different constraints. All examples are performed by MATLAB 7.3 on a personal computer of the Intel Core2 Duo CPU T7250 with 2G memory.

Example 6.1. We test the least squares solutions to Eq. (1.1) with orthogonal constraint by Theorem 2.1. The test matrices A and C in this example are constructed as in (6.1). The singular values are uniformly distributed in the interval $(0, 1)$. The right hand side matrices B and D are constructed by (6.2), where the orthogonal matrix X^* is chosen by

$$[X^*, \mathrm{temp}] = \mathrm{qr}(1 - 2\,\mathrm{rand}(n)).$$

Table 1
Variant matrix sizes n for orthogonal constrained least squares solutions

n	CPU (s)	ϵ	$\|XX^{\mathrm{T}} - I_n\|_F$
100	0.09	$3.72 * 10^{-14}$	$2.27 * 10^{-14}$
200	0.26	$7.05 * 10^{-14}$	$5.71 * 10^{-14}$
300	0.62	$9.88 * 10^{-14}$	$4.10 * 10^{-14}$
500	3.04	$1.62 * 10^{-13}$	$8.99 * 10^{-14}$
700	9.73	$2.24 * 10^{-13}$	$1.29 * 10^{-14}$
1000	29.03	$3.11 * 10^{-13}$	$1.79 * 10^{-14}$

For different n, the residual error ϵ and orthogonal error $\|XX^{\mathrm{T}} - I_n\|_F$ can reach the precision 10^{-13} and 10^{-14}, but both of them seem not to depend on the matrix size n very much. However, the CPU time grows quickly as n increases. In Table 1, we list the CPU time, the residual error ϵ, and orthogonal error $\|XX^{\mathrm{T}} - I_n\|_F$, respectively.

Example 6.2. In this example, we test the efficiency of the our algorithms when the coefficient matrices have different condition numbers. We ask for the least squares solutions to (1.1) with symmetric idempotent constraint by Theorem 4.1. The test matrices A and C with singular values are set as follows:

```
[Ua, temp] = qr(1 − 2*rand(n));
[Va, temp] = qr(1 − 2*rand(n));
da = [1 + rand(9*n/10,1); 10^(−a)*(rand(n/10,1) + 0.1)];
A = Ua*diag(da)*Va′;
[Uc, temp] = qr(1 − 2*rand(n));
[Vc, temp] = qr(1 − 2*rand(n));
dc = [1 + rand(9*n/10,1); 10^(−c)*(rand(n/10,1) + 0.1)];
C = Uc*diag(dc)*Vc′;
```

where $a > 0,\ c > 0$ are two constants that determine the magnitudes of the condition numbers of A and C, respectively. And the matrices $B,\ D$ are constructed by (6.2), where symmetric idempotent matrix X^* satisfies the following rules:

$$x = \mathrm{rand}(n,1) > 0.5, \quad [U_x, \mathrm{temp}] = \mathrm{qr}(1 - 2\,\mathrm{rand}(n)), \quad X^* = U_x\,\mathrm{diag}(x)U_x^{\mathrm{T}}.$$

In Table 2, we list the experiment results for given $n = 500$. The CPU time does not vary significantly with the condition numbers of A and C. The symmetric precision $\|X^{\mathrm{T}} - X\|_F$ is always zero. However, the residual error ϵ and idempotent error $\|X^2 - X\|_F$ can reach 10^{-12} and 10^{-14}, respectively.

Example 6.3. Finally, we test our algorithm (Theorem 5.2) for the P-commuting symmetric orthogonal constraint. The symmetric matrix P with two different eigenvalues is generated as follows:

```
[H, temp] = qr(1 − 2*rand(n));
d = [repmat(1,[1,n/2]),repmat(4,[1,n/2])];
P = Hdiag(d)H′.
```

The coefficient matrices A and C are constructed by (6.1). And the matrices B, D are constructed by (6.2) with

$$X^* = H\,\mathrm{diag}(X_1, X_2)H^{\mathrm{T}},$$

where $X_i,\ i = 1, 2$ are two symmetric orthogonal matrices generated as follows:

```
[Ux1, temp] = qr(1 − 2*rand(n/2));
d1 = 1 − 2*rand(n/2,1);
X1 = Ux1*diag(sign(d1))*Ux1′;
[Ux2, temp] = qr(1 − 2*rand(n/2));
d2 = 1 − 2*rand(n/2,1);
X2 = Ux2*diag(sign(d2))*Ux2′.
```

Table 2
Variant condition numbers for symmetric idempotent least squares solutions, $n = 500$

cond(A)	cond(C)	CPU (s)	ϵ	$\|X^2 - X\|_F$
$1.98 * 10^2$	$1.56 * 10^{14}$	2.49	$1.75 * 10^{-12}$	$6.91 * 10^{-14}$
$1.92 * 10^3$	$1.97 * 10^{13}$	2.60	$1.46 * 10^{-12}$	$5.91 * 10^{-14}$
$1.77 * 10^4$	$1.98 * 10^{12}$	2.46	$1.83 * 10^{-12}$	$7.39 * 10^{-14}$
$1.86 * 10^5$	$1.84 * 10^{11}$	2.56	$1.85 * 10^{-12}$	$7.54 * 10^{-14}$
$1.97 * 10^6$	$1.78 * 10^{10}$	2.59	$1.91 * 10^{-12}$	$7.62 * 10^{-14}$
$1.64 * 10^7$	$1.69 * 10^9$	2.39	$1.88 * 10^{-12}$	$7.59 * 10^{-14}$
$1.16 * 10^8$	$1.99 * 10^8$	2.36	$1.82 * 10^{-12}$	$7.38 * 10^{-14}$
$1.54 * 10^9$	$1.98 * 10^7$	2.56	$1.83 * 10^{-12}$	$6.21 * 10^{-14}$
$1.70 * 10^{10}$	$1.97 * 10^6$	2.50	$1.47 * 10^{-12}$	$6.10 * 10^{-14}$
$1.84 * 10^{11}$	$1.91 * 10^5$	2.45	$1.89 * 10^{-12}$	$5.74 * 10^{-14}$
$1.85 * 10^{12}$	$1.61 * 10^4$	2.42	$1.79 * 10^{-12}$	$7.11 * 10^{-14}$
$1.57 * 10^{13}$	$1.97 * 10^3$	2.54	$1.52 * 10^{-12}$	$7.28 * 10^{-14}$
$1.92 * 10^{14}$	$1.21 * 10^2$	2.49	$1.91 * 10^{-12}$	$6.29 * 10^{-14}$

Table 3
P-commuting symmetric orthogonal least squares solutions

n	CPU (s)	ϵ	$\|X^T - X\|_F$	$\|X^2 - I_n\|_F$	$\|PX - XP\|_F$
100	0.07	$2.54 * 10^{-14}$	$2.36 * 10^{-14}$	$4.34 * 10^{-14}$	$1.45 * 10^{-13}$
200	0.23	$4.37 * 10^{-14}$	$3.24 * 10^{-14}$	$8.17 * 10^{-14}$	$2.58 * 10^{-13}$
400	1.13	$8.54 * 10^{-14}$	$4.22 * 10^{-14}$	$1.66 * 10^{-14}$	$4.81 * 10^{-13}$
600	3.91	$1.26 * 10^{-14}$	$5.37 * 10^{-14}$	$2.47 * 10^{-14}$	$6.65 * 10^{-13}$
800	9.18	$1.67 * 10^{-13}$	$3.56 * 10^{-13}$	$3.26 * 10^{-13}$	$8.56 * 10^{-13}$

We list the experiment results for variant matrix sizes n in Table 3. For different n, the residual error ϵ, symmetric error $\|X^T - X\|_F$, orthogonal error $\|XX^T - I_n\|_F$, and the norm $\|PX - XP\|_F$ all can reach the precision 10^{-13}, but they seem not to depend on the matrix size n very much. In this example, the symmetric error $\|X^T - X\|_F$ is not zero. However, the CPU time grows quickly as n increases.

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